Point forecasts of wind with complex ARIMA: design and results of the first experiment

Wind Forecasting with Complex ARIMA — working notes

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1 Purpose

The paper models wind as a single complex variable, combining speed and direction, with complex ARIMA (cARIMA). This note describes the first forecasting experiment on real data and its results. The experiment asks whether cARIMA gives better point forecasts of 10-minute wind than two simple alternatives:

- *persistence*, which forecasts that the wind stays as it was at the last observation;
- pairs of ordinary (real-valued) ARIMA models, one for each of two components of the wind, estimated independently of each other.

Forecasts are made from 10 minutes to 6 hours ahead. They are evaluated on speed, on direction and on the wind vector as a whole.

The note first describes the data (Section 2) and how the wind is modelled (Section 3), then the design of the forecasting study (Section 4). The results are in two parts: the models estimated on the first window of data (Section 5) and the forecast accuracy (Section 6).

2 Data

- **Source.** The Kaggle "Wind Turbine SCADA" dataset, file `T1.csv`. SCADA (supervisory control and data acquisition) is the monitoring system of a wind turbine; it records measurements at the turbine, here every 10 minutes. The file covers one turbine in Turkey for the whole of 2018 and contains wind speed (m/s) and wind direction (degrees). More details are in `data/README.md`.
- **Missing values.** A complete 10-minute grid for 2018 has 52,560 time points; the file has 50,530 of them. The 2,030 missing points include three gaps longer than 3 days.
- **Preparation** (`R/02_prepare_data.R`):
  - the data are put on the complete 10-minute grid, with the missing points left empty;
  - gaps of at most 36 points (6 hours) are filled by linear interpolation of the two Cartesian coordinates of the wind vector, $\operatorname{Re} Z_t$ and $\operatorname{Im} Z_t$ (defined in Section 3.1);
  - the 10 zero speeds are replaced by 0.1 m/s, so that the logarithm of speed exists.

The 10-minute changes of wind are heavy-tailed: most changes are small, but some are very large. Table 1 gives the kurtosis $\mathrm{E}(x - \mathrm{E}\,x)^4/(\operatorname{Var} x)^2$ of the changes between consecutive observations. It equals 3 for a normal distribution; larger values mean heavier tails. All four changes are far from normal. About 3% of the direction changes exceed 30°, and 0.5% exceed 90°. The changes of the Cartesian coordinates combine changes of speed and of direction, and $\operatorname{Re} Z_t$ has the heaviest tails of all: a sudden turn at high speed moves the wind vector a long way.

Figure 1 shows the same changes as histograms with the counts on a log scale, so that the tails are visible; a normal distribution would appear as a parabola.

- **Left, change of log speed.** The distribution is sharply peaked at zero: half of the changes are within ± 0.06 (about $\pm 6\%$ of the speed), with a standard deviation of 0.16. But the counts decay roughly linearly on the log scale, i.e. exponentially rather than like a normal distribution, and reach changes of -1.6 and $+2.4$ (speeds falling to a fifth or rising elevenfold within 10 minutes). Such large relative changes happen at low speeds: in all 66 changes larger than 1 in absolute value, one of the two speeds is below 3.7 m/s.
- **Middle, change of the recorded angle.** Besides the central peak there are two side peaks near $\pm 2\pi$. They are not real turns of the wind but the artificial jumps that occur when the direction crosses $\pm\pi$ (696 times; Section 3.6). Because of them the standard deviation is 0.71 rad.
- **Right, shortest-arc change of direction.** With the jumps removed, the side peaks disappear and the standard deviation falls to 0.25 rad (14°). The centre is again very narrow (half of the changes are within about $\pm 3°$), and the tails are heavy and extend to $\pm\pi$: there are 63 turns of more than 150° within 10 minutes.

These heavy tails motivate the heavy-tailed error distribution in Section 3.4, and the side peaks motivate the unwrapping of the angle in Section 3.6.

Table 1: Kurtosis of the changes between consecutive 10-minute observations (normal distribution: 3). v_t is speed, θ_t direction and Z_t the wind as a complex number (Section 3.1); wrap$(\cdot)$ is defined in (9).

change	kurtosis
$\log v_t - \log v_{t-1}$	14.4
$\mathrm{wrap}(\theta_t - \theta_{t-1})$	52.8
$\mathrm{Re}\, Z_t - \mathrm{Re}\, Z_{t-1}$	131.9
$\mathrm{Im}\, Z_t - \mathrm{Im}\, Z_{t-1}$	25.5

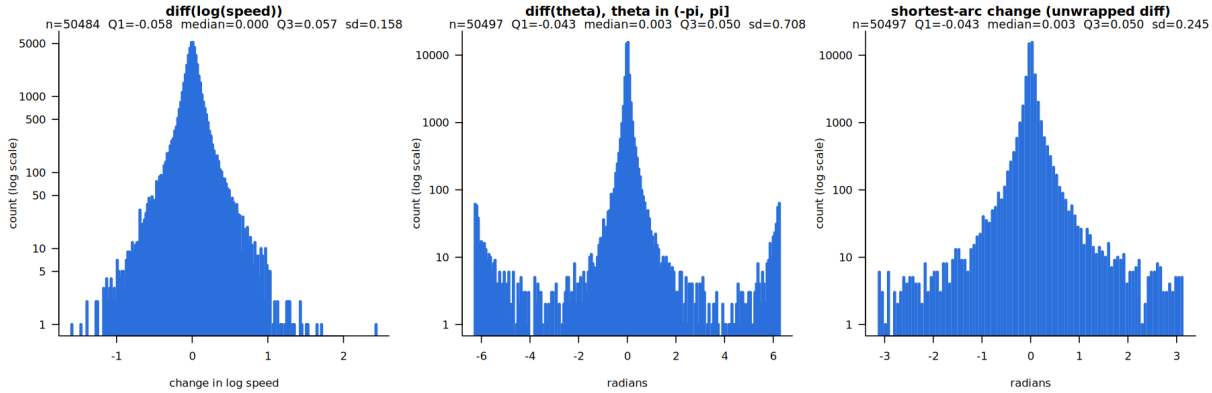


Figure 1: Histograms of the changes between consecutive 10-minute observations, counts on a log scale. Left: change of log speed. Middle: change of the direction angle as recorded in $(-\pi, \pi]$, in radians. Right: shortest-arc change of direction, in radians. The line above each panel gives the number of changes, the quartiles and the standard deviation.

3 How the wind is modelled

3.1 Wind as a complex number

With speed $v_t > 0$ and direction angle $\theta_t \in (-\pi, \pi]$ (in radians), the wind at time t is

$$Z_t = v_t e^{\mathrm{i}\theta_t} = v_t \cos\theta_t + \mathrm{i}\, v_t \sin\theta_t. \tag{1}$$

The modulus $|Z_t|$ is the speed and the argument $\operatorname{Arg} Z_t$ is the direction. $\operatorname{Re} Z_t$ and $\operatorname{Im} Z_t$ are the Cartesian coordinates of the wind vector. They are not the zonal and meridional components used in meteorology, which are measured along the east and north axes; here the angle is simply the recorded direction.

3.2 Complex ARIMA

The cARIMA(p, d, q) model for a complex series y_t is

$$\Big(1 - \sum_{j=1}^{p} \phi_j B^j\Big)(1-B)^d\, y_t = \beta_0 + \Big(1 + \sum_{j=1}^{q} \theta_j B^j\Big)\varepsilon_t, \qquad \phi_j, \theta_j, \beta_0 \in \mathbb{C}, \tag{2}$$

where B is the backshift operator ($By_t = y_{t-1}$), p and q are the autoregressive (AR) and moving average (MA) orders, and d is the order of differencing. With $d = 1$ the constant β_0 is a *drift*: a fixed complex change added at every step. With $d = 0$ it determines the mean of the series, $\beta_0/(1 - \sum_j \phi_j)$.

Multiplying by a complex coefficient $\phi = a + \mathrm{i}b = re^{\mathrm{i}\alpha}$ *scales* by $r = |\phi|$ and *rotates* by $\alpha = \operatorname{Arg}\phi$. For the pair $(\operatorname{Re} y, \operatorname{Im} y)$,

$$\begin{pmatrix} \operatorname{Re}(\phi y) \\ \operatorname{Im}(\phi y) \end{pmatrix} = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \begin{pmatrix} \operatorname{Re} y \\ \operatorname{Im} y \end{pmatrix}, \tag{3}$$

so cARIMA is a bivariate VARIMA whose coefficient matrices are restricted to this form: 2 real parameters per lag instead of 4.

The models are estimated by maximum likelihood with `complex::clm()`. It fits (2) for the series in levels: the observations lost to differencing and lags at the start of the sample are filled in by extrapolation, and the forecasts are produced directly for y_t rather than for its differences.

3.3 The error distribution and non-circularity

In the basic model, the error ε_t is *complex normal*. For a complex random variable with zero mean, two second moments are needed, and we use the terms of Svetunkov and Svetunkov (2024):

- the *conjugate variance* $\sigma^2 = \mathrm{E}(\varepsilon_t \bar{\varepsilon}_t) = \mathrm{E}\,|\varepsilon_t|^2$, computed with the complex conjugate; it is real and measures the overall size of the error;
- the *direct variance* $\varsigma^2 = \mathrm{E}(\varepsilon_t^2)$, computed without the conjugate (elsewhere called the pseudo-variance); it is complex in general and describes the shape of the error cloud in the plane.

Together they give the covariance matrix of the real and imaginary parts:

$$\operatorname{Var}(\operatorname{Re}\varepsilon_t) = \tfrac{1}{2}(\sigma^2 + \operatorname{Re}\varsigma^2), \quad \operatorname{Var}(\operatorname{Im}\varepsilon_t) = \tfrac{1}{2}(\sigma^2 - \operatorname{Re}\varsigma^2), \quad \operatorname{Cov}(\operatorname{Re}\varepsilon_t, \operatorname{Im}\varepsilon_t) = \tfrac{1}{2}\operatorname{Im}\varsigma^2. \tag{4}$$

With $\varsigma^2 = 0$ the error is *circular*: the two parts have the same variance and are uncorrelated, so the cloud of errors is a circle. With $\varsigma^2 \neq 0$ it is *non-circular*: the cloud is an ellipse. For wind, non-circularity means that the changes of the wind vector are larger along some directions than others. We measure it by the ratio of the two variances,

$$\kappa = \frac{|\varsigma^2|}{\sigma^2} \in [0, 1], \tag{5}$$

which is 0 for a circular error and 1 for an error that always lies on one line. A pair of independent real models, one for $\operatorname{Re} Z_t$ and one for $\operatorname{Im} Z_t$, ignores the correlation between the two errors.

3.4 The complex generalised normal distribution

Because the changes of wind are heavy-tailed (Table 1), the models are also estimated with a *complex generalised normal* error (Novey et al., 2010; `distribution = "dcgnorm"` in `clm()`). For the error $e = z - \mu$, with conjugate scale $\alpha^2 > 0$, direct scale $\psi^2 \in \mathbb{C}$ ($|\psi^2| < \alpha^2$) and shape $\beta > 0$, the density is

$$f(z) = \frac{2\beta}{\pi\,\Gamma(1/\beta)\,2^{1/\beta}\sqrt{\alpha^4 - |\psi^2|^2}}\exp\Big(-\tfrac{1}{2}Q^{\beta}\Big), \qquad Q = \frac{2(\alpha^2|e|^2 - \operatorname{Re}(\overline{\psi^2}\,e^2))}{\alpha^4 - |\psi^2|^2}. \tag{6}$$

- $\beta = 1$ gives the complex normal distribution with conjugate variance α^2 and direct variance ψ^2. A shape $\beta < 1$ gives heavier tails ($\beta = 0.5$ is similar to the Laplace distribution); $\beta > 1$ gives lighter tails.
- For other β, the conjugate and direct variances equal α^2 and ψ^2 multiplied by $2^{1/\beta}\Gamma(2/\beta)/(2\Gamma(1/\beta))$. In `dcgnorm()` the two scales are the arguments `scaleConjugate` and `scaleDirect`.
- The shape β is estimated together with the ARIMA coefficients. For any given values of these, the maximum likelihood estimates of the two scales have a closed form, so they are substituted into the likelihood ("concentrated out") and not searched for numerically.

With heavy tails, a few very large changes can pull the estimates of a normal model, in particular the drift. A heavy-tailed likelihood gives these observations less weight.

3.5 Why logarithms

Besides Z_t itself, the models are applied to the complex logarithm of the wind,

$$\log Z_t = \log v_t + \mathrm{i}\,\tilde{\theta}_t, \tag{7}$$

where $\tilde{\theta}_t$ is the unwrapped direction angle (Section 3.6).

- The logarithm *separates speed and direction*: the real part is log speed and the imaginary part is the angle. A cARIMA for $\log Z_t$ is still one complex model: the two parts are linked through the complex coefficients and the error covariance.
- Changes of speed are roughly *proportional* to the speed (larger when it is windy), so log speed is a more natural scale than speed.
- Forecasts are transformed back by $\hat{Z} = \exp(\widehat{\log Z})$, which always gives a positive speed and a valid direction.
- $\exp(\widehat{\log Z})$ is not the expected value of Z; no correction for this is made.
- The drift of a model for $\log Z_t$ with $d = 1$ has a direct meaning: $\operatorname{Re}\beta_0$ is a steady percentage change of speed, and $\operatorname{Im}\beta_0$ is a steady rotation of the direction, in radians per 10 minutes. There is no physical reason for the wind to keep turning in one direction for weeks, so models for $\log Z_t$ are also fitted without the constant.

3.6 Differences and unwrapping of the angle

- At 10-minute resolution the wind is very persistent: over several weeks its level wanders without returning to a stable mean. The models are therefore fitted mainly to changes,

$d = 1$. In logarithms the differences have a clear meaning:

$$(1 - B) \log Z_t = \log \frac{v_t}{v_{t-1}} + \mathrm{i}\,(\tilde{\theta}_t - \tilde{\theta}_{t-1}), \tag{8}$$

the log ratio of consecutive speeds plus i times the change of direction.

- The recorded angle jumps by about 2π whenever the direction crosses $\pm\pi$. For example, a turn from 179° to −179° is a change of 2°, but the recorded values differ by 358°. This happens 696 times in `T1.csv`. In differences these jumps would appear as huge artificial shocks (Figure 1, middle).
- The angle is therefore *unwrapped* within each estimation window $t = 1, \ldots, T$:

$$\tilde{\theta}_1 = \theta_1, \qquad \tilde{\theta}_t = \tilde{\theta}_{t-1} + \mathrm{wrap}(\theta_t - \theta_{t-1}), \qquad \mathrm{wrap}(x) = \mathrm{atan2}(\sin x, \cos x) \in (-\pi, \pi]. \tag{9}$$

  $\mathrm{wrap}(\cdot)$ returns the equivalent angle in $(-\pi, \pi]$, so each step of $\tilde{\theta}_t$ is the *shortest-arc* change of direction, and $\tilde{\theta}_t$ can leave $(-\pi, \pi]$. Forecasts of the direction are wrapped back into $(-\pi, \pi]$ for evaluation.
- The unwrapped angle is used in $\log Z_t$ and in the real benchmark models for the angle (Section 4.2). Z_t itself needs no unwrapping: it is continuous when the direction crosses $\pm\pi$, because cos and sin are periodic.
- For Z_t, $d = 0$ is also a plausible choice: the wind vector could revert to a mean. One set of models chooses $d \in \{0, 1\}$ automatically (Section 4.2).

4 Design of the forecasting study

4.1 Rolling origin

The forecasts are evaluated with a *rolling origin*: the models are repeatedly estimated on a window of past data and forecast the next 6 hours, for many different ends of the window.

- A *forecast origin* is the last observation used for estimation. The estimation window is the 4 weeks up to the origin ($T = 4032$ observations), and the forecast *horizon* is $h = 1, \ldots, 36$ steps ahead (10 minutes to 6 hours).
- An origin is used only if its window has no missing values and its 6-hour horizon has no missing values and no values from interpolations longer than 3 points (30 minutes). So the forecasts are compared with values that were actually observed, or nearly so.
- The first window that satisfies this (2018-01-30 14:40 to 2018-02-27 14:30) is the *selection window*. It is used only to choose the model orders (Section 4.2).
- 300 forecast origins are spread evenly over all later usable origins, from 2018-02-27 14:40 to 2018-12-31 17:50. They are spread evenly over the times of day, but not over the months: months with more gaps have fewer usable origins. There are about 40 origins in each month from March to September (except June, with 5), none in October, 14 in November and 27 in December (plus 2 at the end of February).
- At every origin, the parameters of all models are estimated again on that origin's window; only the orders stay fixed.

4.2 Choice of model orders

The orders are chosen once, on the selection window, and kept for all 300 origins.

- **cARIMA with $d = 1$ (grid search).** All models with $p, q \in \{0, \ldots, 3\}$ are estimated, and the one with the lowest AICc is chosen. AICc is Akaike's information criterion with a correction for small samples: it balances the fit (log-likelihood) against the number of parameters.
- **cARIMA with automatic choice of d and of the constant** (`auto.carima()` in the `complex` package):

  ```
  auto.carima(y, orders = list(ar = 3, i = 1, ma = 3), constant = NULL).
  ```

  For each $d \in \{0, 1\}$ it runs a *stepwise* search: starting from cARIMA$(0, d, 0)$, it fits all models whose p and q differ by at most one from the current model, moves to the best of them, and stops when AICc no longer improves ($p, q \leq 3$). The constant is first included for $d = 0$ and excluded for $d = 1$; the best model for each d is then also tried with the opposite choice, and the search continues from there. Models with different d lose different numbers of initial observations, so AICc is computed on a common sample that drops the first few observations for all of them.
- **Starting values.** The default starting values of `clm()` are the complex least squares estimates for the differenced series, with $0.1 + 0.1\mathrm{i}$ for each MA coefficient. From them, `clm()` sometimes stops at a poor local maximum of the likelihood; for example, a larger model ending with a lower likelihood than a smaller model nested in it. Therefore every cARIMA model is estimated from several starting values, and the fit with the highest likelihood is kept:
  - in the grid search, the default and the estimates of the nested models $(p - 1, q)$ and $(p, q - 1)$, with zeros for the new coefficient;
  - in `auto.carima()`, the Hannan–Rissanen estimates and the estimates of the closest model already fitted, with zeros for the new coefficients. The Hannan–Rissanen method regresses the series on its own lags and on the lags of errors approximated by a long autoregression, by complex least squares; it gives consistent estimates of the AR and MA coefficients without numerical optimisation;
  - in the rolling origin, the default and the estimates from the selection window.
- **Real benchmarks.** `forecast::auto.arima()` with $d = 1$ (a stepwise search of the same kind for a real series, by AICc), and `smooth::auto.msarima()`. The latter also chooses d itself, so it is applied to the differenced series, and the chosen model is then fitted to the series with $d = 1$.

4.3 Models compared

Table 2 lists the models. Each row gives the series that is modelled, the error distribution, how the orders are chosen, and whether the model has a constant. The rows are grouped into persistence, cARIMA with $d = 1$, cARIMA with automatic choice of d, and the pairs of real models. The names are built as follows:

- `carima_`: cARIMA with $d = 1$ and orders from the grid search; `auto_carima_`: cARIMA with d, the orders and the constant chosen by `auto.carima()`;
- `logZ` or `z`: the model is for $\log Z_t$ or for Z_t;
- `_cgn`: complex generalised normal error (otherwise complex normal);
- `_nd`: no drift, i.e. no constant β_0;
- `arima_` and `msarima_`: two independent real ARIMA models, from `auto.arima()` and `auto.msarima()` respectively, for the pair of series named after the prefix. Their forecasts

are combined into $\hat{Z}$: `speed_angle` (speed and unwrapped angle), `logSpeed_angle` (log speed and unwrapped angle) and `re_im` ($\mathrm{Re}\, Z_t$ and $\mathrm{Im}\, Z_t$). A negative speed forecast is set to zero.

Table 2: Models compared. The errors of the cARIMA models are complex normal or complex generalised normal; those of the real benchmarks are normal.

model	series	error	orders	constant β_0
`persistence`	–	–	$\hat{Z}_{t+h\|t} = Z_t$ for all h	–
`carima_logZ`	$\log Z$	normal	$d = 1$, grid search	yes
`carima_logZ_cgn`	$\log Z$	gen. normal	$d = 1$, grid search	yes
`carima_logZ_nd`	$\log Z$	normal	$d = 1$, grid search	no
`carima_logZ_cgn_nd`	$\log Z$	gen. normal	$d = 1$, grid search	no
`carima_z`	Z	normal	$d = 1$, grid search	yes
`carima_z_cgn`	Z	gen. normal	$d = 1$, grid search	yes
`auto_carima_logZ`	$\log Z$	normal	`auto.carima()`	chosen
`auto_carima_logZ_cgn`	$\log Z$	gen. normal	`auto.carima()`	chosen
`auto_carima_z`	Z	normal	`auto.carima()`	chosen
`auto_carima_z_cgn`	Z	gen. normal	`auto.carima()`	chosen
`arima_*`, `msarima_*`	two real series	normal	`auto.arima()`, `auto.msarima()`; $d = 1$	chosen

4.4 Error measures

For origin t and horizon h, with forecast $\hat{Z}_{t+h|t}$ and actual value Z_{t+h}, three errors are computed in the original units:

$$e_{t,h}^{\text{speed}} = |\hat{Z}_{t+h|t}| - |Z_{t+h}| \quad \text{(m/s)}, \tag{10}$$

$$e_{t,h}^{\text{angle}} = \tfrac{180}{\pi}\,\mathrm{wrap}(\mathrm{Arg}\,\hat{Z}_{t+h|t} - \mathrm{Arg}\, Z_{t+h}) \quad \text{(degrees)}, \tag{11}$$

$$e_{t,h}^{\text{vector}} = |\hat{Z}_{t+h|t} - Z_{t+h}| \quad \text{(m/s)}. \tag{12}$$

The angle error is the shortest-arc difference, between $-180°$ and $180°$. The vector error is the length of the difference between the forecast and the actual wind vectors, so it combines errors of speed and direction. Each measure is summarised by the root mean squared error over the 300 origins, $\mathrm{RMSE}_h = \left(\frac{1}{300}\sum_t e_{t,h}^2\right)^{1/2}$, for each horizon, and over all 36 horizons together (column "$h = 1$–36").

5 Estimated models

This section describes the models chosen and estimated on the selection window. In the rolling origin the parameters are estimated again at every origin, so these values illustrate the models rather than being used for the forecasts. Standard errors are not reported.

Table 3 gives, for each cARIMA model, the chosen orders (p, d, q), the estimated complex coefficients, the shape of the generalised normal ("–" for the normal models), and the log-likelihood. Each complex coefficient has two real parameters: its real part acts on the same component (e.g. log speed on log speed), its imaginary part couples the two components. Log-likelihoods can be compared between models for the same series, but not between Z_t and $\log Z_t$, which are on different scales. The main points are:

Table 3: cARIMA models estimated on the selection window: orders, constant $\hat{\beta}_0$, AR coefficient $\hat{\phi}_1$, MA coefficients $\hat{\theta}_j$ (signs as in (2)), shape $\hat{\beta}$ of the generalised normal, and the log-likelihood.

model	(p,d,q)	$\hat{\beta}_0$	$\hat{\phi}_1$	$\hat{\theta}_1,\ldots,\hat{\theta}_q$	$\hat{\beta}$	log-likelihood
`carima_logZ`	(0,1,3)	$-0.0003 + 0.0122$i	–	$-0.0856 - 0.0148$i $-0.0511 - 0.0018$i $-0.0250 + 0.0011$i	–	493.33
`carima_logZ_cgn`	(0,1,2)	$0.0001 + 0.0012$i	–	$0.0607 + 0.0056$i $-0.0649 + 0.0047$i	0.212	5510.52
`carima_z`	(0,1,2)	$-0.0026 - 0.0005$i	–	$-0.0281 - 0.0245$i $-0.1250 + 0.0375$i	–	−9906.10
`carima_z_cgn`	(0,1,2)	$-0.0016 + 0.0088$i	–	$0.0306 + 0.0023$i $-0.1000 + 0.0250$i	0.408	−8420.80
`carima_logZ_nd`	(1,1,2)	none	$0.6264 - 0.0111$i	$-0.7074 + 0.0000$i $0.0000 + 0.0000$i	–	492.51
`carima_logZ_cgn_nd`	(0,1,2)	none	–	$0.0650 + 0.0089$i $-0.0774 + 0.0131$i	0.212	5509.75
`auto_carima_z`	(1,0,2)	$-0.0159 + 0.0138$i	$0.9930 + 0.0022$i	$-0.0270 - 0.0287$i $-0.1293 + 0.0394$i	–	−9888.61
`auto_carima_z_cgn`	(1,0,2)	$-0.0102 + 0.0212$i	$0.9943 + 0.0006$i	$0.0344 + 0.0010$i $-0.0918 + 0.0197$i	0.409	−8408.73
`auto_carima_logZ`	(1,1,1)	$0.0000 + 0.0044$i	$0.6427 + 0.0044$i	$-0.7250 - 0.0145$i	–	499.72
`auto_carima_logZ_cgn`	(0,1,2)	none	–	$0.0685 + 0.0017$i $-0.0710 + 0.0003$i	0.213	5509.95

- **Orders.** All models are small: at most one AR and three MA terms. With d chosen automatically, the models for Z_t have $d = 0$ and those for $\log Z_t$ have $d = 1$. `auto_carima_logZ_cgn` is the same specification as `carima_logZ_cgn_nd`: $(0, 1, 2)$ without a constant.
- **Heavy tails.** The estimated shape of the generalised normal is $\hat{\beta} \approx 0.21$ for $\log Z_t$ and $\hat{\beta} \approx 0.41$ for Z_t, far below the normal value of 1. The generalised normal raises the log-likelihood by about 5,000 for $\log Z_t$ and 1,500 for Z_t.
- **Drift of the direction.** For $\log Z_t$ with a normal error, the imaginary part of the constant is 0.0122 rad (0.70°) per 10 minutes in `carima_logZ`, a rotation of about 25° over the 6-hour horizon. The real model for the angle (`arima_angle`, Table 5) has the same drift, 0.0121. With the generalised normal the drift is ten times smaller, 0.0012. `auto.carima()` keeps the constant for $\log Z_t$ with the normal error and drops it with the generalised normal.
- Z_t **with** $d = 0$**.** $\hat{\phi}_1 \approx 0.993$–$0.994$, with an argument close to zero: the model is close to a random walk and does not rotate. A deviation from the mean halves in $\log 0.5 / \log|\hat{\phi}_1| \approx$ 100–120 steps, i.e. 16–20 hours (the *half-life*). The implied mean wind vector $\beta_0/(1-\phi_1)$ is $-2.63 + 1.14$i (normal) and $-2.16 + 3.49$i (generalised normal), 2.9 and 4.1 m/s long. The mean *speed* in the selection window is 8.5 m/s, but the mean *vector* is only 3.5 m/s long, because winds from different directions cancel. So at long horizons these forecasts move towards a short vector, i.e. a low speed.
- **A sub-optimal fit in the order choice.** Refitting `carima_logZ_nd`, $(1, 1, 2)$, gives a log-likelihood of 492.51, higher than the 489.91 found in the grid search. The best fit has $\hat{\theta}_2 = 0$ exactly, which suggests that the optimiser stayed at the starting values taken from the nested $(1, 1, 1)$ model. The grid search for this model thus compared a sub-optimal fit, and $(1, 1, 1)$ may have been the better choice. All other models reproduce the grid-search log-likelihood to within 0.02.

Table 4 describes the residuals of the same models on the selection window: the standard deviations of their real and imaginary parts, the correlation between the two, and the non-circularity (5). κ is estimated by the ratio of the sample direct and conjugate variances of the centred residuals $\tilde{\varepsilon}_t = \hat{\varepsilon}_t - \frac{1}{T}\sum_s \hat{\varepsilon}_s$: $\hat{\kappa} = |\sum_t \tilde{\varepsilon}_t^2| / \sum_t |\tilde{\varepsilon}_t|^2$.

- The values are almost the same for all models of the same series: the residuals depend on the series much more than on the orders or the distribution.
- $\hat{\kappa}$ is 0.27–0.33 for all models, so the errors are clearly non-circular: the direct variance is about a third of the conjugate variance.
- For Z_t, the two coordinates have different spreads (0.90 and 0.80 m/s) and are positively correlated (0.31), so the error ellipse is tilted: the wind vector changes most along a diagonal of the plane, i.e. along a prevailing direction.
- For $\log Z_t$, the imaginary part (direction) has a larger standard deviation, 0.26 rad or about 15°, than the real part (log speed), 0.20, and the two are almost uncorrelated. Here the non-circularity comes mainly from the different spreads of speed and direction.

Table 4: Residuals on the selection window: standard deviations and correlation of the real and imaginary parts, and the non-circularity $\hat{\kappa}$.

model	sd $\mathrm{Re}\,\hat{\varepsilon}$	sd $\mathrm{Im}\,\hat{\varepsilon}$	cor($\mathrm{Re}\,\hat{\varepsilon}, \mathrm{Im}\,\hat{\varepsilon}$)	$\hat{\kappa}$
`carima_logZ`	0.196	0.265	−0.076	0.304
`carima_logZ_cgn`	0.201	0.264	−0.054	0.271
`carima_z`	0.901	0.798	0.310	0.330
`carima_z_cgn`	0.904	0.800	0.311	0.332
`carima_logZ_nd`	0.195	0.265	−0.075	0.306
`carima_logZ_cgn_nd`	0.201	0.264	−0.051	0.269
`auto_carima_z`	0.900	0.796	0.310	0.331
`auto_carima_z_cgn`	0.903	0.799	0.313	0.334
`auto_carima_logZ`	0.195	0.266	−0.075	0.306
`auto_carima_logZ_cgn`	0.201	0.264	−0.053	0.269

Table 5 gives the real models used in the benchmark pairs, also estimated on the selection window. Each pair in Table 2 combines two of them; for example, `arima_re_im` uses `arima_re` and `arima_im`.

- The models for speed are larger than any cARIMA model: `arima_speed` has 5 AR and 3 MA terms, with large coefficients of opposite signs that nearly cancel.
- Both models for the angle have the same drift, 0.0121 rad per 10 minutes, as the normal cARIMA for $\log Z_t$. All pairs that contain the angle (`speed_angle`, `logSpeed_angle`) therefore rotate their forecasts at this rate.
- The models for $\mathrm{Re}\,Z_t$ and $\mathrm{Im}\,Z_t$ are small and have no or a tiny drift.

Table 5: Real benchmark models estimated on the selection window. Each is for one series: speed, log speed, unwrapped angle, $\operatorname{Re} Z$ or $\operatorname{Im} Z$. ϕ_j: AR coefficients, θ_j: MA coefficients (signs as in (2)), c: drift.

model	(p, d, q)	drift	coefficients
`arima_speed`	(5,1,3)	no	$\phi_1 = -0.5156$, $\phi_2 = 0.1343$, $\phi_3 = 0.7397$, $\phi_4 = 0.0330$, $\phi_5 = 0.0443$, $\theta_1 = 0.4364$, $\theta_2 = -0.2834$, $\theta_3 = -0.7965$
`msarima_speed`	(3,1,2)	yes	$\phi_1 = 0.3978$, $\phi_2 = 0.2726$, $\phi_3 = 0.0589$, $\theta_1 = -0.4786$, $\theta_2 = -0.3482$, $c = -0.0002$
`arima_logSpeed`	(1,1,2)	no	$\phi_1 = 0.7313$, $\theta_1 = -0.9495$, $\theta_2 = 0.0767$
`msarima_logSpeed`	(3,1,3)	yes	$\phi_1 = 0.2165$, $\phi_2 = 0.1815$, $\phi_3 = 0.1305$, $\theta_1 = -0.4353$, $\theta_2 = -0.2192$, $\theta_3 = -0.1209$, $c = -0.0001$
`arima_angle`	(2,1,0)	yes	$\phi_1 = 0.0346$, $\phi_2 = -0.0267$, $c = 0.0121$
`msarima_angle`	(0,1,1)	yes	$\theta_1 = 0.0355$, $c = 0.0121$
`arima_re`	(2,1,1)	no	$\phi_1 = -0.3312$, $\phi_2 = -0.1251$, $\theta_1 = 0.2956$
`msarima_re`	(0,1,2)	yes	$\theta_1 = -0.0360$, $\theta_2 = -0.1247$, $c = -0.0022$
`arima_im`	(2,1,2)	no	$\phi_1 = 0.1092$, $\phi_2 = -0.0453$, $\theta_1 = -0.1265$, $\theta_2 = -0.1186$
`msarima_im`	(0,1,2)	yes	$\theta_1 = -0.0196$, $\theta_2 = -0.1664$, $c = -0.0004$

6 Forecast accuracy

Tables 6–8 give the RMSE over the 300 origins at selected horizons; all 36 horizons are in `results/tables/rmse_speed.csv`, `rmse_angle.csv` and `rmse_z.csv`. Each table has one row per model, grouped as in Table 2, and one column per horizon; the last column pools all 36 horizons. Bold marks the lowest value in each column. All models produced forecasts at all origins.

Speed (Table 6). The errors grow from about 0.71 m/s at 10 minutes to 3.0–3.4 m/s at 6 hours, and the differences between models are small compared with this growth.

- At $h = 1$ and $h = 3$ the cARIMA models for $\log Z$ with the generalised normal error are best (0.706 and 1.045 m/s), slightly ahead of persistence (0.710 and 1.048).
- From $h = 6$ on, the real pair `arima_speed_angle`, which models speed directly, is best; over all horizons its RMSE is 2.214 m/s against 2.282 for persistence.
- The models for Z with $d = 0$ (`auto_carima_z`, `auto_carima_z_cgn`) are the worst at long horizons (3.354 and 3.183 at $h = 36$). As explained in Section 5, their forecasts move towards the mean wind vector, which is much shorter than the typical wind speed.

Table 6: RMSE of speed $|\hat{Z}|$ (m/s).

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 24$	$h = 36$	$h = 1$–36
`persistence`	0.710	1.048	1.571	1.995	2.607	3.082	2.282
`carima_logZ`	0.715	1.061	1.575	2.000	2.602	3.085	2.283
`carima_logZ_cgn`	**0.706**	1.047	1.575	2.014	2.634	3.129	2.309
`carima_logZ_nd`	0.714	1.058	1.576	1.996	2.593	3.071	2.276
`carima_logZ_cgn_nd`	0.706	**1.045**	1.569	1.991	2.609	3.083	2.284
`carima_z`	0.718	1.055	1.569	1.989	2.598	3.080	2.281
`carima_z_cgn`	0.716	1.054	1.574	1.999	2.613	3.093	2.294
`auto_carima_logZ`	0.715	1.067	1.582	2.013	2.637	3.178	2.321
`auto_carima_logZ_cgn`	0.707	1.047	1.570	1.992	2.608	3.083	2.284
`auto_carima_z`	0.726	1.083	1.615	2.105	2.818	3.354	2.466
`auto_carima_z_cgn`	0.720	1.062	1.584	2.031	2.690	3.183	2.352
`arima_logSpeed_angle`	0.727	1.083	1.598	2.014	2.586	3.023	2.271
`arima_re_im`	0.727	1.054	1.569	1.978	2.576	3.044	2.256
`arima_speed_angle`	0.720	1.047	**1.557**	**1.951**	**2.500**	**2.981**	**2.214**
`msarima_logSpeed_angle`	0.728	1.087	1.607	2.022	2.596	3.049	2.283
`msarima_re_im`	0.728	1.061	1.574	1.996	2.603	3.085	2.283
`msarima_speed_angle`	0.717	1.051	1.567	1.976	2.557	3.033	2.247

Direction (Table 7). The errors grow from about 12.4° at 10 minutes to 46–50° at 6 hours.

- Over all horizons, `carima_logZ_nd` (log Z, normal error, no constant) is best, 38.21° against 38.59° for persistence, and it is best at 22 of the 36 individual horizons.
- The models that rotate the direction by a drift are the worst at long horizons: the normal cARIMA for $\log Z$ with a constant (`carima_logZ`, `auto_carima_logZ`) and all real pairs that contain the angle model, at about 49.5° at $h = 36$ and 40.2–40.4° over all horizons. The pairs `arima_speed_angle` and `arima_logSpeed_angle` have identical rows because they use the same angle model (and likewise for `msarima_`); they differ only in speed.
- With the generalised normal error the drift is small, and the corresponding models (`carima_logZ_cgn`, `auto_carima_logZ_cgn`) are close to persistence.

Table 7: RMSE of direction (degrees).

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 24$	$h = 36$	$h = 1$–36
`persistence`	12.42	20.74	26.78	36.49	43.01	46.14	38.59
`carima_logZ`	12.67	20.71	26.73	36.99	45.42	49.47	40.18
`carima_logZ_cgn`	12.50	20.88	26.82	36.57	43.41	46.31	38.74
`carima_logZ_nd`	12.64	**20.44**	**26.29**	**36.14**	42.63	**45.77**	**38.21**
`carima_logZ_cgn_nd`	12.46	20.90	26.84	36.60	43.02	46.13	38.62
`carima_z`	12.81	20.96	26.82	36.53	42.82	45.94	38.50
`carima_z_cgn`	12.66	21.00	26.96	36.27	**42.57**	46.10	38.40
`auto_carima_logZ`	12.52	20.83	27.04	37.24	45.58	49.57	40.29
`auto_carima_logZ_cgn`	12.48	20.87	26.82	36.55	42.99	46.12	38.59
`auto_carima_z`	12.98	21.47	27.64	36.63	44.31	48.28	39.80
`auto_carima_z_cgn`	12.68	21.13	27.13	36.43	43.02	46.80	38.75
`arima_logSpeed_angle`	12.49	21.09	27.15	37.27	45.64	49.42	40.38
`arima_re_im`	12.87	21.14	27.06	36.75	42.60	45.84	38.57
`arima_speed_angle`	12.49	21.09	27.15	37.27	45.64	49.42	40.38
`msarima_logSpeed_angle`	**12.38**	21.15	27.29	37.27	45.68	49.43	40.40
`msarima_re_im`	12.74	20.85	26.78	36.58	42.94	45.99	38.56
`msarima_speed_angle`	**12.38**	21.15	27.29	37.27	45.68	49.43	40.40

Wind vector (Table 8). The vector error combines speed and direction. It grows from about 0.96 m/s at 10 minutes to 4.5–5.5 m/s at 6 hours.

- At $h = 1$ and $h = 3$, cARIMA for $\log Z$ with the generalised normal error is best (0.955 and 1.620 m/s, against 0.960 and 1.634 for persistence). At $h = 6$, `arima_re_im` and `carima_logZ_cgn_nd` are equal best (2.294).
- From $h = 12$ on, `auto_carima_z_cgn` (Z, $d = 0$, generalised normal) is best, and its advantage grows with the horizon: 4.501 against 4.721 for persistence at $h = 36$, and 3.455 against 3.567 over all horizons. Moving towards the mean wind vector costs speed accuracy (Table 6) but reduces the vector error, because when the direction is uncertain, a shorter forecast vector is closer on average to the actual one.
- The models with a rotating drift are the worst here too: `carima_logZ`, `auto_carima_logZ` and the real pairs with the angle model have 3.90–3.99 over all horizons, about 10% worse than persistence.

Table 8: RMSE of the wind vector $|\hat{Z} - Z|$ (m/s).

model	$h = 1$	$h = 3$	$h = 6$	$h = 12$	$h = 24$	$h = 36$	$h = 1$–36
`persistence`	0.960	1.634	2.307	2.996	4.104	4.721	3.567
`carima_logZ`	0.965	1.658	2.363	3.139	4.507	5.385	3.915
`carima_logZ_cgn`	**0.955**	**1.620**	2.298	2.998	4.126	4.763	3.584
`carima_logZ_nd`	0.961	1.631	2.297	2.991	4.078	4.700	3.550
`carima_logZ_cgn_nd`	0.955	1.620	2.294	2.981	4.093	4.710	3.555
`carima_z`	0.964	1.630	2.296	2.979	4.077	4.699	3.549
`carima_z_cgn`	0.963	1.635	2.306	2.984	4.091	4.715	3.560
`auto_carima_logZ`	0.965	1.676	2.388	3.186	4.580	5.535	3.987
`auto_carima_logZ_cgn`	0.955	1.622	2.296	2.983	4.093	4.711	3.556
`auto_carima_z`	0.967	1.633	2.307	2.985	4.030	4.587	3.509
`auto_carima_z_cgn`	0.964	1.634	2.299	**2.946**	**3.977**	**4.501**	**3.455**
`arima_logSpeed_angle`	0.971	1.684	2.390	3.150	4.488	5.360	3.909
`arima_re_im`	0.974	1.631	**2.294**	2.969	4.045	4.648	3.522
`arima_speed_angle`	0.968	1.669	2.374	3.132	4.468	5.360	3.900
`msarima_logSpeed_angle`	0.973	1.693	2.406	3.162	4.500	5.380	3.923
`msarima_re_im`	0.973	1.645	2.309	2.995	4.098	4.723	3.567
`msarima_speed_angle`	0.965	1.673	2.378	3.132	4.480	5.365	3.901

7 Findings

1. **No model beats persistence on all three measures.** The gains over persistence are small, at most about 3% over all horizons (vector: 3.455 vs 3.567), as is typical for 10-minute wind forecast from its own past only. Poorly specified models are up to about 10% worse than persistence.

2. **Short horizons (h = 1–3):** cARIMA for $\log Z$ with the generalised normal error (`carima_logZ_cgn`, `carima_logZ_cgn_nd`) is best for the vector error and for speed.

3. **The drift rotates the forecasts.** With a normal error, the constant of a model for $\log Z$ rotates the forecast direction at a steady rate. `carima_logZ` and `auto_carima_logZ` are among the worst models for direction (about 40° against 38.6° for persistence over all horizons) and for the vector error. Without the constant, `carima_logZ_nd` has the best direction forecasts of all models (38.21°). The generalised normal largely removes the problem even with a constant, because its drift estimate is much smaller.

4. **Z with $d = 0$:** `auto_carima_z_cgn` has the lowest vector error of all models over all horizons (3.455, against 3.567 for persistence and 3.522 for the best pair of real models, `arima_re_im`). Its advantage comes from $h = 12$ onwards. It is worse for speed (2.352 against 2.282), because its forecasts move towards the short mean wind vector. `auto_carima_z`, with the normal error, behaves the same way and has the worst speed forecasts (2.466).

5. **Speed alone** is forecast best by `arima_speed_angle` from $h = 6$ onwards (2.214 over all horizons), the real model fitted to speed directly.

6. **Heavy tails matter:** $\hat{\beta} \approx 0.2$ for $\log Z$ and 0.4 for Z, and the generalised normal helps most where the drift of the normal model distorts the forecasts.

8 Limitations and next steps

- Only point forecasts are evaluated: no predictive distributions, no tests of the differences in accuracy (e.g. Diebold–Mariano), and no other error measures (e.g. MAE).
- One turbine and one year. The orders are chosen on a single window in February and kept for the whole year.
- The forecasts for $\log Z$ are transformed back without a correction for the bias of $\exp(\cdot)$.
- The order of `carima_logZ_nd` was chosen from a sub-optimal fit (Section 5).
- Not yet included: daily cycles (e.g. Fourier terms), VAR and VARMA benchmarks, which are needed to test the argument that cARIMA is more parsimonious, forecasts of turbine power, and other datasets.

9 Reproducing the results

```
Rscript R/02_prepare_data.R              # needs data/raw/T1.csv
Rscript R/03_select_orders.R [model ...] # no arguments = all models
Rscript R/04_rolling_origin.R [model ...]
Rscript R/05_evaluate.R
Rscript R/06_parameters.R
```

`data/README.md` explains how to download `T1.csv`. Shared code is in `R/functions.R`. `03_select_orders.R` chooses the orders on the selection window, `04_rolling_origin.R` produces the forecasts, `05_evaluate.R` computes the RMSE tables, and `06_parameters.R` produces Tables 3–5. Scripts `03` and `04` can be run for selected models only; their results are merged with the existing ones. On 4 cores, the order choice and the 300 forecasts of the four `auto_carima` models took 23 minutes and 2.8 hours.

References

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